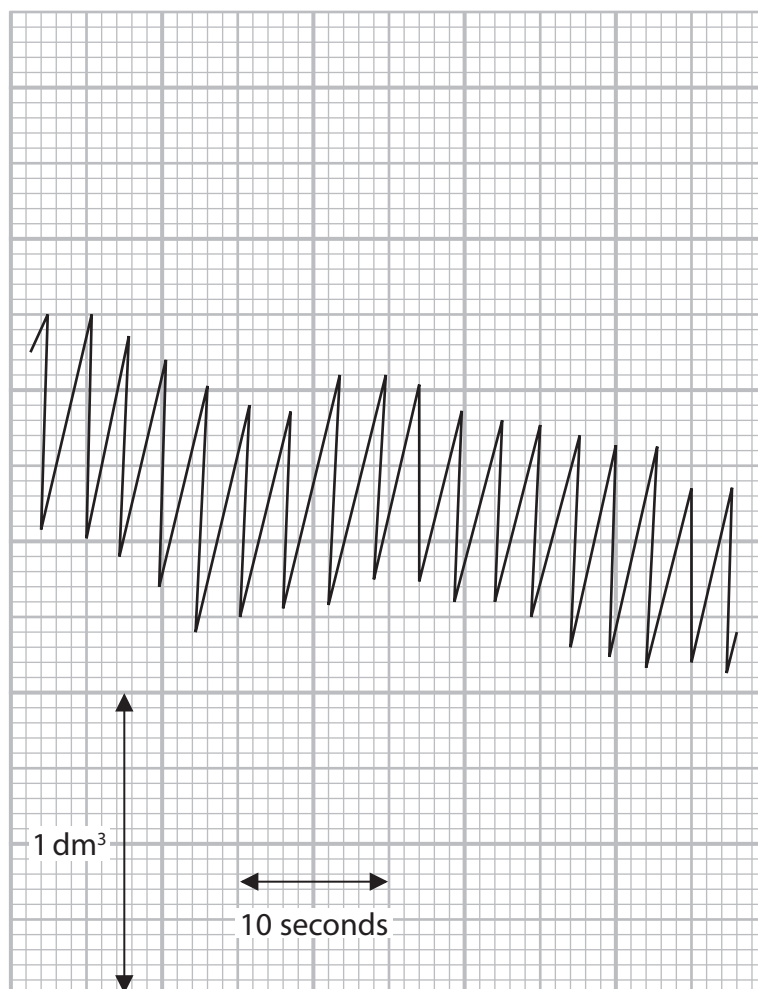


- 2 Information about depth and rate of breathing in humans can be obtained using a spirometer.

A student investigated the effect of exercise on breathing rate and respiratory minute ventilation.

The diagram shows a spirometer trace for a person resting quietly.



- (a) (i) Calculate the mean breathing rate for this person.

(1)

..... breaths  $\text{min}^{-1}$

(ii) Estimate the mean respiratory minute ventilation for this person.

(2)

.....  $\text{dm}^3 \text{ min}^{-1}$

(iii) During exercise, respiratory minute ventilation increases.

Explain why it is necessary for the respiratory minute ventilation to increase during exercise.

(2)

.....

.....

.....

.....

.....

.....

(b) (i) State **one** abiotic and **one** biotic variable that could affect this investigation.

(2)

Abiotic variable

Biotic variable

(ii) Choose **one** of the variables you have identified in (b)(i).

Describe how this variable could be controlled and the effect it could have on the results if it is not controlled.

(2)

Variable .....

Describe how this variable is controlled.

Describe the effect it could have on the results if it is not controlled.

**(Total for Question 2 = 9 marks)**